

EDUCATION	School of Computer Science, Shanghai Jiao Tong University Shanghai, China <i>Ph.D. in Computer Science and Engineering</i> 2022 - 2027 (<i>expected</i>) • Advisor: Prof. Junchi Yan • Research area: Science of Large Language Models, Theory of Deep Learning
	UM-SJTU Joint Institute, Shanghai Jiao Tong University Shanghai, China <i>B.E. in Electrical and Computer Engineering</i> 2018 - 2022 • Top 5%, Minor in Data Science.
EXPERIENCES	Tencent Hunyuan Pretraining Group. Beijing, China 2026/05 - present • Research Intern, with Qingyun Talent Program.
	ByteDance Seed Edge. Beijing, China 2025/12 - 2026/05 • Research Intern, focusing on optimizer and hyperparameter transfer in pretraining.
	MiniMax Inc. Shanghai, China 2025/04 - 2025/11 • Research Intern, contributing to MiniMax-M2 model.
	National Institute of Informatics. Tokyo, Japan 2023/09 - 2024/03 • Research Intern, advised by Prof. Mahito Sugiyama.
PUBLICATIONS (CORE)	(* indicates equal contributions; † indicates correspondence.) 1. Zhanpeng Zhou*†, Yuhan Sun*, Bingrui Li, Jinbo Wang, Huaijin Wu, Lei Wu, Junchi Yan. Sharp-to-Flat Early, Noise-Governed Late: Local Geometry in Language Model Pre-Training. In Submission. 2. Zhanpeng Zhou*, Wenjie Zhou, Yufei Gu, Juqiu Wang, Guanwei Zhang, Ruobing Xie, Xianbiao Qi, Yushi Bai, Kaifeng Lyu, Shunyu Yao. From LR to ELR: A Better Heuristic for Pretraining. Hunyuan Research. 3. Zhanpeng Zhou*†, Mingze Wang*, Yuchen Mao, Bingrui Li, Junchi Yan†. Sharpness-Aware Minimization Efficiently Selects Flatter Minima Late in Training. ICLR 2025 (Spotlight). 4. Zhanpeng Zhou*, Zijun Chen*, Yilan Chen, Bo Zhang, Junchi Yan. On the Emergence of Cross-Task Linearity in the Pretraining-Finetuning Paradigm. ICML 2024. 5. Zhanpeng Zhou*, Wen Shen*, Huixin Chen*, Ling Tang, Quanshi Zhang. Batch Normalization Is Blind to the First and Second Derivatives of the Loss. AAAI 2024 (Oral). 6. Zhanpeng Zhou, Yongyi Yang, Xiaojiang Yang, Junchi Yan, Wei Hu. Going Beyond Linear Mode Connectivity: The Layerwise Linear Feature Connectivity. NeurIPS 2023. 7. Jianhao Huang*, Zhanpeng Zhou*, Renqiu Xia, Baharan Mirzasoleiman, Weijie Su, Wei Huang. How Transformers Learn to Plan via Multi-Token Prediction. COLM 2026. 8. Tongcheng Zhang*, Zhanpeng Zhou*, Mingze Wang, Andi Han, Wei Huang, Taiji Suzuki, Junchi Yan. On the Learning Dynamics of Two-Layer Networks with Label Noise SGD. AAAI 2026 (Oral). 9. Jinbo Wang*, Mingze Wang*, Zhanpeng Zhou*, Junchi Yan, Weinan E, Lei Wu. The Sharpness Disparity Principle in Transformers for Accelerating Language Model Pre-Training. ICML 2025. 10. Tongtian Zhu, Tianyu Zhang, Mingze Wang, Zhanpeng Zhou†, Can Wang. On the Surprising Effectiveness of a Single Global Merging in Decentralized Learning. ICLR 2026 (Oral).

PUBLICATIONS (OTHERS)	(* indicates equal contributions; † indicates correspondence.)	
	1. Bingrui Li, Jiaxin Wen, Zhanpeng Zhou, Jun Zhu, Jianfei Chen. Efficient Hyperparameter Tuning via Trajectory Invariance Principle. In Submission. 2. Jinbo Wang, Binghui Li, Zhanpeng Zhou, Mingze Wang, Yuxuan Sun, Jiaqi Zhang, Xunliang Cai, Lei Wu. Fast Catch-Up, Late Switching: Optimal Batch Size Scheduling via Functional Scaling Laws. ICLR 2026. 3. Xuan Cui, Yunfei Zhao, Huiyue Li, Run Zeng, Jinrui Qian, Wei Duan†, Bo Liu†, Zhanpeng Zhou†. IGU-LoRA: Adaptive Rank Allocation via Integrated Gradients and Uncertainty-Aware Scoring. ICLR 2026. 4. Andi Han*†, Wei Huang*†, Zhanpeng Zhou*†, Gang Niu, Wuyang Chen, Junchi Yan, Akiko Takeda, Taiji Suzuki. On the Role of Label Noise in the Feature Learning Process. ICML 2025. 5. Zijun Chen*, Zhanpeng Zhou*, Bo Zhang, Weinan Zhang, Xi Sun, Junchi Yan. SE-Merging: A Self-Enhanced Approach for Dynamic Model Merging. IJCNN 2025. 6. Zhanpeng Zhou, Yongyi Yang, Mahito Sugiyama, Junchi Yan. New Evidence of the Two-Phase Learning Dynamics of Neural Networks. ICML 2025 Workshop. 7. Zhanpeng Zhou, Yongyi Yang, Jie Ren, Mahito Sugiyama, Junchi Yan. On the Cone Effect in the Learning Dynamics. ICLR 2025 Workshop. 8. Bingrui Li, Wei Huang, Andi Han, Zhanpeng Zhou, Taiji Suzuki, Jun Zhu, Jianfei Chen. On the Optimization and Generalization of Two-layer Transformers with Sign Gradient Descent. ICLR 2025 (Spotlight). 9. Yiting Chen, Zhanpeng Zhou, Junchi Yan. Going Beyond Neural Network Feature Similarity: The Network Feature Complexity and Its Interpretation Using Category Theory. ICLR 2024.	
PROJECTS	PMuon Preconditioning Muon with Gradient Covariance Before the Polar Step • SOTA on Track 3 Optimization of modded-nanogpt, tying NorMuon. MiniMax-M2 Series Mini Activations Unleashing Max Real-World Intelligence • Early verification on Muon optimizers and blockwise LR methods.	
AWARDS	• Top Reviewer Award, NeurIPS 2025 2025/10 • National Scholarship (top 0.2%), Ministry of Education 2024/11 • Top Internship Evaluation, National Institute of Informatics 2024/03 • Outstanding Graduate Student, Shanghai Jiao Tong University, 2022/05 • Yu Liming Scholarship, Shanghai Jiao Tong University 2021/11 • John Wu & Jane Sun Scholarship, Shanghai Jiao Tong University 2020/11 • Best Technology Award in Summer Expo, Shanghai Jiao Tong University 2019/08	
SERVICES	Area Chair : CPAL '26, ICML '26 FoGen, ICLR '26 DeLTa Conference Reviewer : ICML ('22-25), NeurIPS ('22-25), ICLR ('24-26) Journal Reviewer : T-PAMI, Intelligent Computing (Science Partner)	